

PATRICE IRADUKUNDA

Statistician, Data Scientist, ML Engineer & AI Engineer

☎ +250 788 850 439 | ✉ patriceir27@gmail.com | 🌐 patriceprodatascientist.netlify.app
🌐 linkedin.com/in/patrice-iradukunda | 🏠 github.com/patriceir27-ship-it | 📍 Kigali, Rwanda

PROFILE

Data Scientist Analyst with a strong foundation in statistics and applied mathematics, specialising in machine learning, predictive modelling, data engineering, and decision intelligence. Experienced in transforming complex, multi-source datasets into analytical products that support real-world decisions across healthcare, research, and business environments. Holds an M.Sc. in Mathematical Sciences (Data Science) from AIMS Rwanda and a B.Sc. in Statistics Applied to Economy with Honours. Interested in applying statistical and machine-learning methods to financial risk, customer intelligence, forecasting, and data-driven business decisions.

EDUCATION

M.Sc. in Mathematical Science *Specialisation: Data Science*
African Institute for Mathematical Sciences (AIMS Rwanda)

Sept 2025 – June 2026
📍 Kigali, Rwanda

- Core modules: Statistical Learning, Deep Learning, Stochastic Modeling, Optimisation, Probability Theory, Linear Algebra, Scientific Computing, Big Data Analysis, Database Systems Management
- Research focus: AI-Assisted Code Generation for Embedded Systems with Computer Vision Applications

B.Sc. in Statistics Applied to Economy *With Honours (A0)*
INES Ruhengeri (Institut d'Enseignement Supérieur de Ruhengeri)

Nov 2020 – Oct 2024
📍 Musanze, Rwanda

- **Thesis:** Assessing the Effects of Climate Change on Agricultural Productivity Across Rwanda's Provinces and Kigali City (*published see Publications*)
- Key courses: Econometrics, Multivariate Analysis, Sampling Methods, Operations Research, Actuarial Science, R Programming
- **Award:** Top Student, Department of Statistics Applied to Economy (2021–22)

RESEARCH & PUBLICATIONS

- [1] Iradukunda, P., Uwilingiyimana, C., & Ushize Rutikanga, J. (2024). Assessing the Effects of Climate Change on Agriculture Productivity Across Rwanda's Provinces and Kigali City. *African Journal of Applied Statistics*, **11**, 1553–1576. doi:10.16929/ajas/2024.1553.282
- [2] Uwilingiyimana, C., Iradukunda, P., Izerimana, L., Hakizimana, J.V., & Yadufashije, C. (2026). Impact of Industrial Attachment Program and Collaboration between Rwandan Industry-Educational Institutions. *Indian Journal of Science and Technology*, **19**(5), 313–321.

PROFESSIONAL EXPERIENCE

Data Scientist
eFiche

Jul 2026 – Present
📍 Kigali, Rwanda

- Developing an automated healthcare market intelligence and business development engine that transforms fragmented healthcare data into actionable business intelligence.
- Building data pipelines integrating national health-facility registries, OpenStreetMap, Google Places, and other sources to discover, clean, standardise, geocode, and deduplicate healthcare facilities.
- Applying data quality, geospatial analytics, facility segmentation, and prioritisation techniques to identify high-potential facilities, market gaps, and business development opportunities.
- Developing an interactive intelligence dashboard combining facility discovery, market analytics, geographic insights, and prioritised opportunities to support business decision-making.

Data Scientist Intern
Prodigy InfoTech

Dec 2025 – June 2026
📍 Remote

- Developed end-to-end data science and machine learning projects, applying Python, SQL, and statistical techniques to real-world datasets.

- Built and evaluated machine learning models for classification and prediction tasks, applying data preprocessing, feature engineering, model training, and performance evaluation.
- Conducted exploratory data analysis and created visualisations and reports to communicate patterns, findings, and model results clearly.
- Strengthened practical skills in Python, SQL, machine learning, data analysis, and data storytelling through project-based deliverables.

Tutorial Assistant

INES Ruhengeri

Dec 2024 – Aug 2025

📍 Musanze, Rwanda

- Supported lecturers in lesson preparation, classroom delivery, and monitoring of student progress across statistics and mathematics courses.
- Administered coursework logistics and provided additional instruction to students requiring academic support.

Internet Researcher & Content Creator

Kawah Coffee Ltd

May – Sept 2024

📍 London, UK (Remote)

- Designed data collection pipelines; cleaned, stored, and managed datasets; analysed market trends to support informed business decisions.
- Produced website and social-media content; monitored company AI tool deployment.

Research Assistant & Data Analyst

Conservation Heritage Turambe

Feb – April 2024

📍 Musanze, Rwanda

- Supported the full research cycle: methodology design, questionnaire development and digitisation, enumerator training, and data quality control.
- Analysed data and produced visualisations and reports for stakeholder communication.

District Super Admin LODA National Census

Musanze District Headquarters

Summer 2023

📍 Musanze, Rwanda

- Trained enumerators on standardised data collection procedures.
- Managed the national census system and ensured data integrity throughout.

Enumerator

Bisate Health Center

Jul–Sept 2021 & Jul–Sept 2022

📍 Musanze, Rwanda

- Collected and entered data into the national health facilities information system.

TECHNICAL SKILLS

Programming	Python, R, SQL, Shell / Bash Scripting
Machine Learning & AI	Scikit-learn, Spark MLlib, TensorFlow / Keras, Supervised & Unsupervised Learning, NLP, Time-Series Analysis, Predictive Modelling, Dimensionality Reduction
Big Data & Cloud	PySpark, ETL Pipelines, AWS (introductory), GCP (introductory), Linux / Bash, Git / GitHub
Visualisation	Power BI, Tableau, Matplotlib, Seaborn
Statistics	Statistical Modelling, Regression Analysis, Probability Theory, Linear Algebra, Optimisation, Sampling Methods
Tools	Jupyter Notebook, VS Code, L ^A T _E X, Microsoft Office Suite

PROJECTS

1. Healthcare Market Intelligence & Business Development Engine(Ongoing)

[GitHub](#)

Automated healthcare market intelligence system integrating multiple facility data sources, data cleaning and deduplication, geospatial analysis, facility prioritisation, market-gap detection, and interactive business intelligence dashboards.

2. AI-Assisted Code Generation for Embedded Systems

[GitHub](#)

Research project developing a hardware-aware AI framework for generating embedded C/C++ and Python computer-vision programs, with resource profiling for execution time, memory, flash usage, and energy consumption.

3. Decision Tree Classifier for Customer Purchase

[GitHub](#)

Machine learning project using demographic and behavioural data to predict customer purchase behaviour through classification modelling.

4. AI-Powered HR Screening & Candidate Recommendation System

[GitHub](#)

AI-powered HR platform that screens and ranks candidates against configurable job-specific criteria, automatically shortlists qualified candidates, and uses NLP to identify skill gaps and generate candidate development recommenda-

tions. Implemented weighted evaluation across skills, experience, education, project relevance, and availability using Google Gemini, with ranked results, explainable score breakdowns, and HR decision-support insights.

5. Global Population Dashboard

 [GitHub](#)

Interactive Power BI dashboard analysing population trends, demographics, and growth patterns across countries.

6. Sentiment Analysis & Public Opinion Mining

 [GitHub](#)

Natural Language Processing project applying text analysis and sentiment classification techniques to extract insights from public opinion data.

AWARDS & CERTIFICATIONS

Nov 2022	Top Student Award , <i>INES-Ruhengeri</i> . Awarded for outstanding academic performance in the Department of Statistics Applied to Economy, 2021–22 academic year.
May 2023	Leadership Mentorship Training , <i>INA-Leader (Online)</i> . Certificate for completing an online leadership development programme.
Jul 2023	African Mathematics School Certificate , <i>INES Ruhengeri</i> . Certificate of attendance at a competitive pan-African mathematics programme.
May 2024	AI, Virtual Assistance & Digital Marketing , <i>INNOV8POD / Remote Skills Academy (Online)</i> . Short-course certification covering applied AI tools and digital marketing strategy.

REFERENCES

Dr. Charline Uwilingiyimana

Deputy Vice-Chancellor,
Academic & Research
INES Ruhengeri

ucharline@ines.ac.rw
+250 788 284 814

Dr. Emmanuel Hagenimana

Senior Lecturer &
Postgraduate Coordinator
INES Ruhengeri

ehagenimana@ines.ac.rw
+250 788 344 997

Mrs. Michaelle Kubwimana

CEO, Kawah Coffee Ltd
Founder, Rwanda Business UK Ltd
London, UK

kubwimana78@gmail.com
+44 7765 749 265